BITS F451- Autonomous Mobile Robotics

Two wheeled self balancing robot

Group 6-Team Members:

- Srujan Deolasee (2019A7PS1139P)

Project Idea and Milestones:

- Travelling a path using a self-balancing two wheeled robot.
- Planned milestones:
  - Make the robot model [done]
  - Use necessary sensors [done]
  - Implement a controller [done]
  - Develop a path planner (static obstacle avoidance) [done]
  - Integrating path planner and the controller [done]

Specifications:

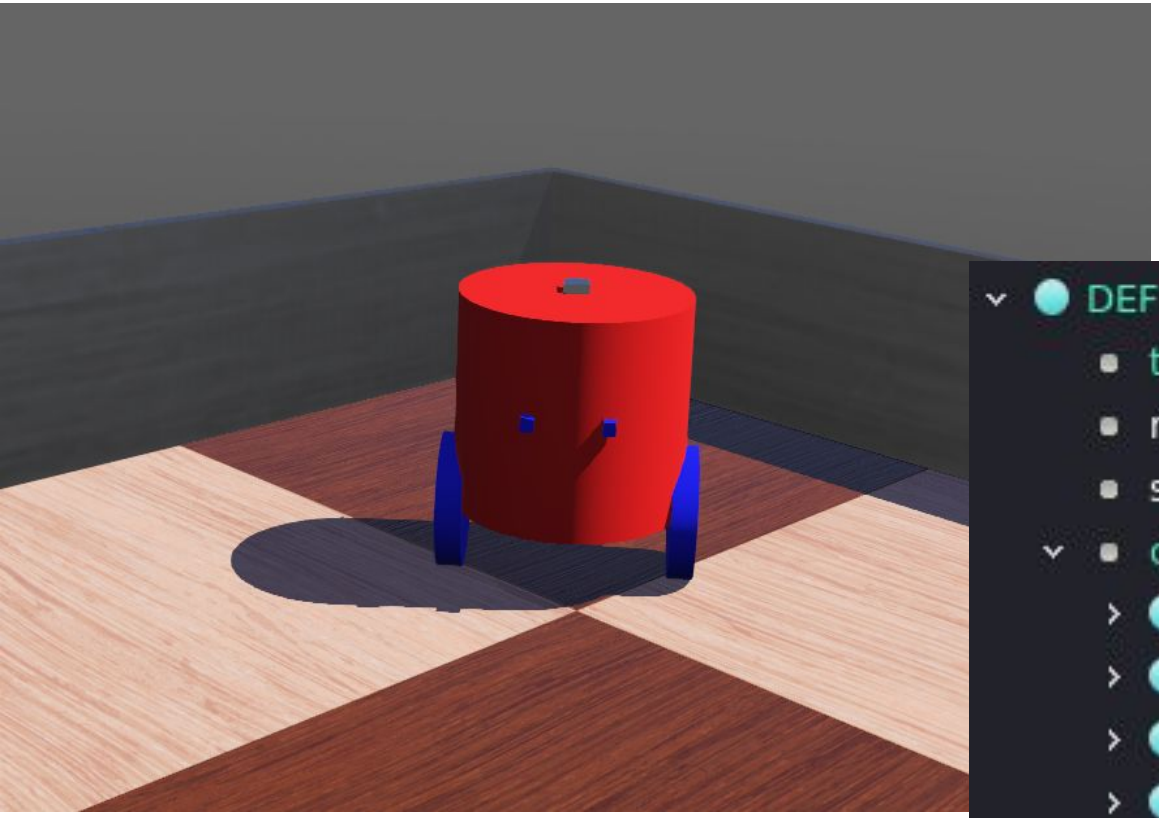
Robot Model:

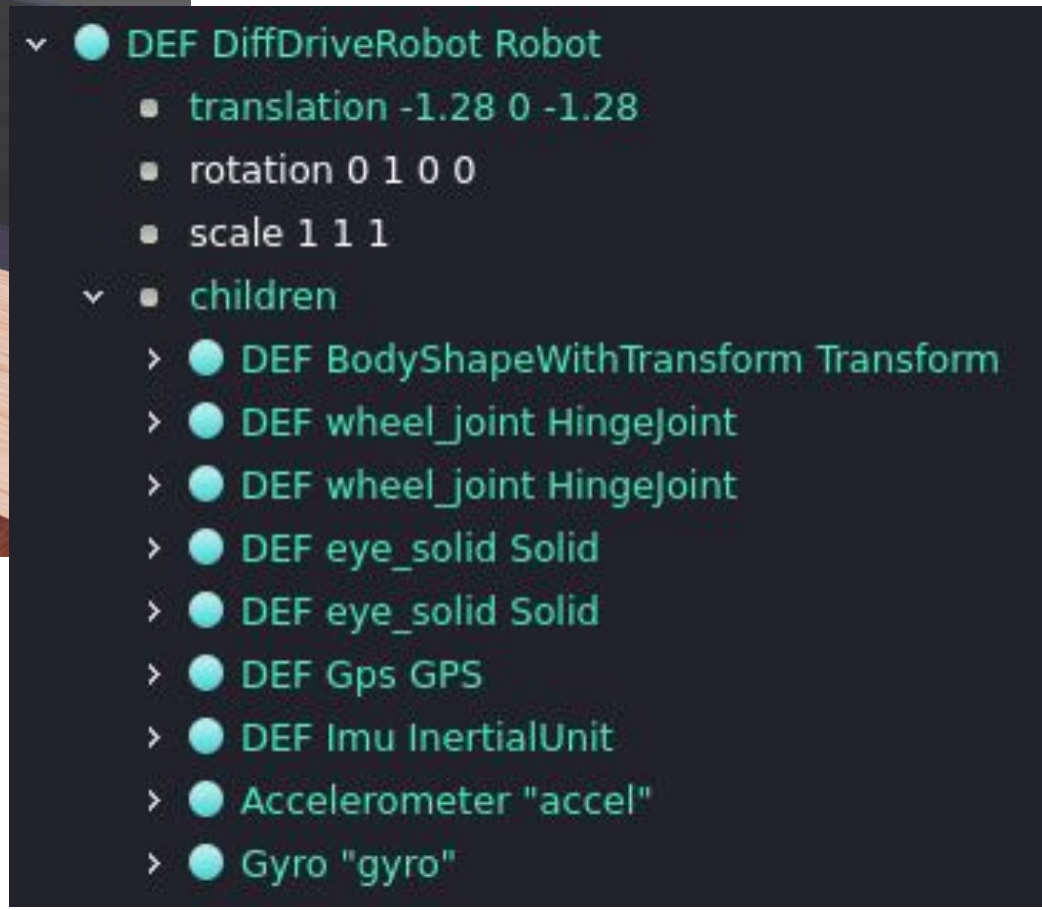
- Used a differential drive robot (simple model)

Sensors:

- IMU (for yaw)
- GPS (for position)
- Wheel Encoders (used in controller)
- Gyro and Accelerometer (used separately as webots gives roll-pitch-yaw readings for IMU)

Robot Model




DEF DiffDriveRobot Robot
translation -1.28 0 -1.28
rotation 0 1 0 0
scale 1 1 1
children
DEF BodyShapeWithTransform Transform
DEF wheel_joint HingeJoint
DEF wheel_joint HingeJoint
DEF eye_solid Solid
DEF eye_solid Solid
DEF Gps GPS
DEF Imu InertialUnit
Accelerometer "accel"
Gyro "gyro"

Specifications:

Controller:

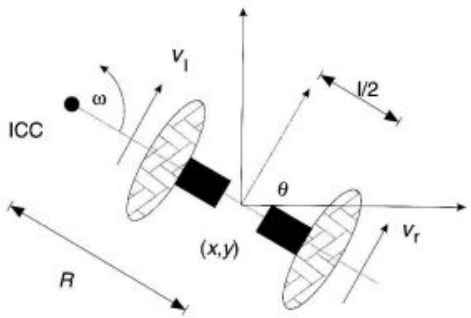
- Linear Quadratic Regulator (LQR)
  - Tuned by trial and error
  - Uses total rotation of wheels, rate of rotation of the wheels, tilt angle calculated by accelerometer and gyrometer, and the gyro reading to calculate the the rotational speed of the wheels for the robot to balance on the spot.

Tilt angle calculation

- Accelerometer- measures all forces/accelerations on the object including the forces that drive the system (motor).
- While in motion, it's affected by vibrations
    - false angle computations even when there is no rotation (High frequency noise).
- gyroscope- measurement is not susceptible to external forces
    - tendency to drift, not returning to zero when the system comes back to original position (Low frequency noise)
- **complementary filter**- removes both of these noises.
- low pass filter and a high pass filter is combined to fuse the pitch angles calculated by the accelerometer and gyroscope readings
    - $\theta = \alpha \cdot \text{gyro rate} \cdot dt + (1 - \alpha) \cdot \text{accel angle}$

Controller (cntd):

- Calculate speed of the wheels needed for the robot to turn or travel in a straight line based on the waypoint
- For turning, rotations of the wheels will be in opposite directions
- For traveling straight, rotation of the wheels will be in the same direction

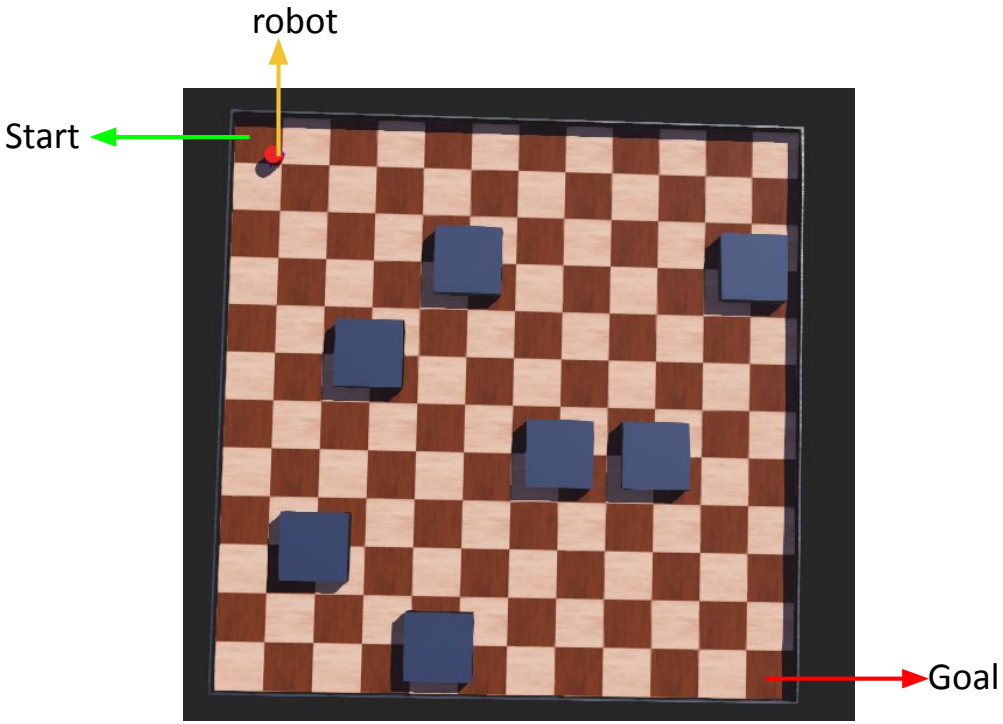


$$\omega \cdot (R + \frac{l}{2}) = V_r$$

$$\omega \cdot (R - \frac{l}{2}) = V_l$$

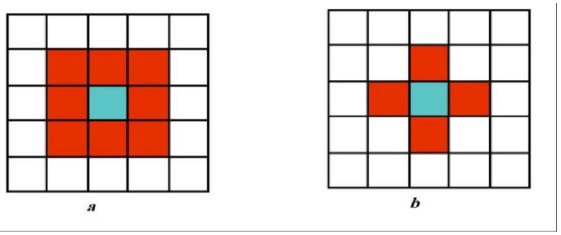
Path Planner

- Map Representation
  - Discrete squares/grid
  - Static obstacles inserted (cuboids)

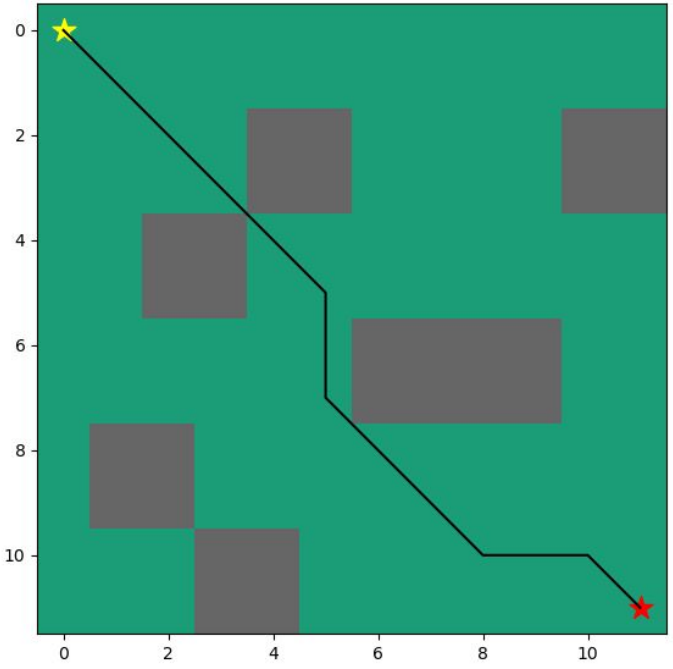


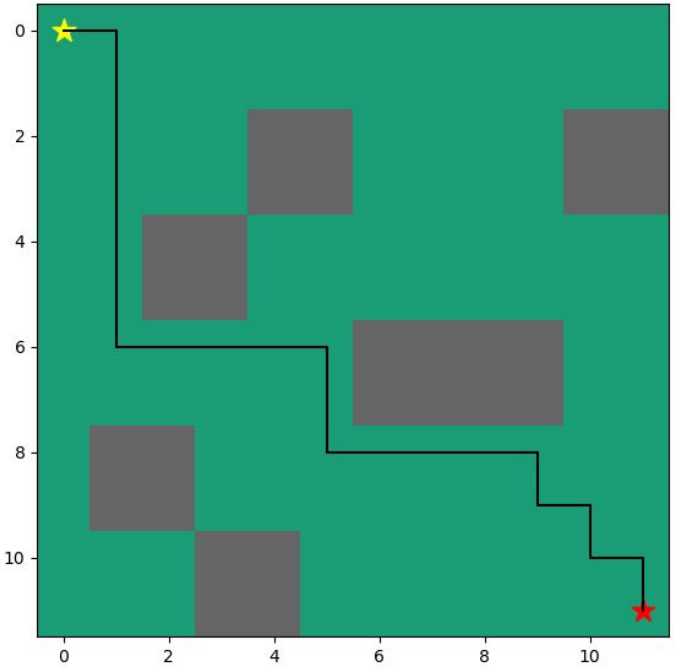
Path Planner

- Implemented A* Path Planner
  - Path depends on number of neighbors:



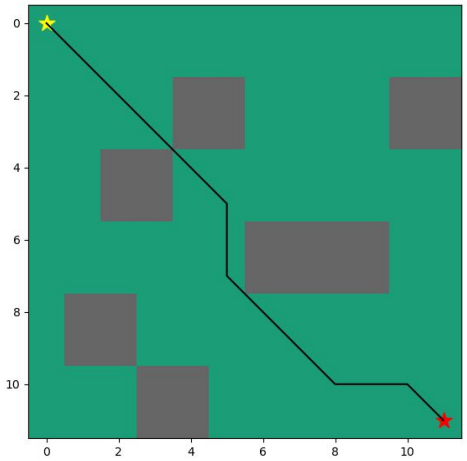
- Moore's neighborhood (a)
- Von Neumann neighborhood (b)

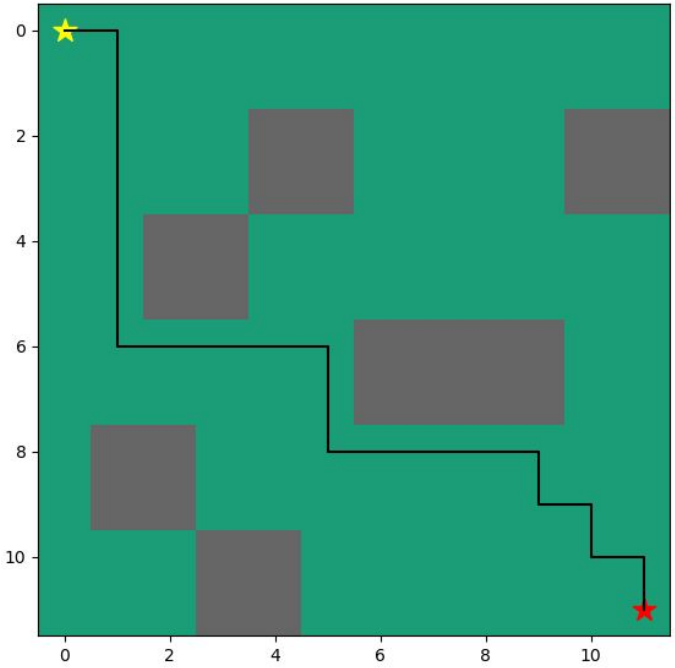


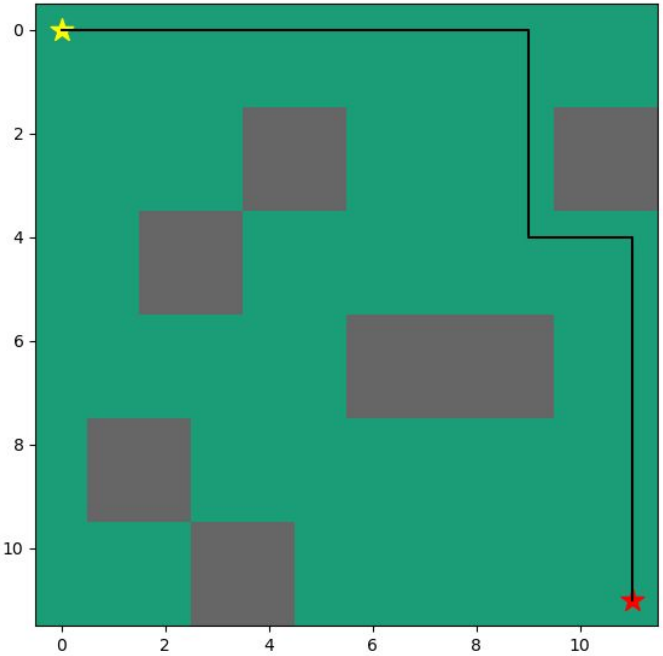


Path Planner

- Implemented A* Path Planner
  - Path depends on heuristic
    - Euclidean
    - Manhattan







Simulation

Future Work/ Improvements

- Developing a more robust and multi-loop control architecture, which will speed up the process of path traversal.
- Dynamic obstacle avoidance can be implemented with the help of sensors like camera or LiDAR.

Thank You